

# Tom Gallagher

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## Education

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### University of Canterbury - Academic Requirements Completed

Bachelor of Engineering (Honours), Software Engineering

Final Year GPA: 6.88

## Skills

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**Soft Skills:** Collaboration, Leadership, Learning From Feedback, Time Management, Adaptability, Critical Thinking, Initiative, Empathy.

**Languages:** Java, Python, JavaScript, C, SQL, HTML, CSS, Prolog.

**Frameworks/Libraries:** ReactJS, Spring Boot, Material UI, Junit, Dask, Cucumber, Mockito,

**Technologies:** Git, AWS ec2, NGINX, Microsoft Azure, JetBrains IDEs, VSCode, GitHub Copilot, Generative AI.

## Projects

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- **Chess Engine (Personal Project)** Java engine using minimax with alpha-beta pruning and bitboards. Integrated with JavaFX GUI and deployed via Spring Boot on AWS EC2. Online version built with React.
- **GreenThumbs (SENG302 Team Project)**  
Agile-built Spring Boot app connecting gardening workers to job posters. Used Java, Thymeleaf, MariaDB, Azure API, JUnit, and Cucumber.
- **Workout Recommender System (DATA301)**  
Suggested new activities using similar users based on heart rate similarity with big data processing.
- **Petition Manager (SENG365)**  
Full-stack web app with REST API (Node.js, MySQL) and React frontend using Material UI.
- **Final-Year Research Project (SENG402)**  
Enabling Real-Time Data Collection, Immediate Model Testing and End-User Training in Prosthetic Hand Control Systems

## Experience & References

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### OssAbility - Intern Software Engineer

Nov 2025 - Present

- Developing a critical Python extension for 3D Slicer to facilitate AI-driven surgical planning for complex orthopedic procedures.
- Integrating the extension with existing machine learning models to enable automated feature identification in patient scans.
- Engineering and testing new AI models to improve the precision and efficiency of the surgical planning workflow.
- Collaborating with the clinical team to refine feature requirements and ensure the tool meets clinical standards

References are available on request